Q1) Design and implement Selenium environment setup to configure WebDriver and verify browser.

```
from selenium import webdriver

# open browser
driver = webdriver.Chrome()

# open website
driver.get("https://www.google.com")

# print title
print(driver.title)

# close browser
driver.quit()
```

Q2) Design and implement basic browser navigation to automate opening, refreshing, and navigating pages.

```
from selenium import webdriver
import time

driver = webdriver.Chrome()

driver.get("https://www.google.com")
time.sleep(2)

driver.get("https://www.facebook.com")
time.sleep(2)

driver.back()
```

```
time.sleep(2)


driver.forward()

time.sleep(2)


driver.refresh()


driver.quit()
```

Q3) Design and implement locators to find web elements using ID, XPath, or CSS selectors.

```
from selenium import webdriver

from selenium.webdriver.common.by import By


driver = webdriver.Chrome()

driver.get("https://www.google.com")


# locate search box using name

search = driver.find_element(By.NAME, "q")

search.send_keys("Selenium")


driver.quit()
```

Q4) Design and implement form automation to fill and submit forms with test data.

```
from selenium import webdriver

from selenium.webdriver.common.by import By


driver = webdriver.Chrome()

driver.get("https://demo.guru99.com/test/newtours/register.php")
```

```
driver.find_element(By.NAME, "firstName").send_keys("Om")

driver.find_element(By.NAME, "lastName").send_keys("Sai")


driver.quit()
```

Q5) Design and implement dropdown and checkbox handling using Select and click methods.

```
from selenium import webdriver

from selenium.webdriver.common.by import By

from selenium.webdriver.support.ui import Select


driver = webdriver.Chrome()

driver.get("https://www.w3schools.com/tags/tryit.asp?filename=tryhtml_select")


driver.switch_to.frame("iframeResult")


dropdown = Select(driver.find_element(By.TAG_NAME, "select"))

dropdown.select_by_index(1)


driver.quit()
```

Q6) Design and implement alert and pop-up handling to interact with JavaScript alerts.

```
from selenium import webdriver

from selenium.webdriver.common.by import By


driver = webdriver.Chrome()

driver.get("https://www.w3schools.com/js/tryit.asp?filename=tryjs_alert")


driver.switch_to.frame("iframeResult")
```

```
driver.find_element(By.TAG_NAME, "button").click()


alert = driver.switch_to.alert

alert.accept()


driver.quit()
```

Q7) Design and implement screenshot capturing to save images of web pages during testing.

```
from selenium import webdriver


driver = webdriver.Chrome()

driver.get("https://www.google.com")


driver.save_screenshot("screenshot.png")


driver.quit()
```

Q8) Design and implement mouse and keyboard actions using the Actions class for advanced interactions.

```
from selenium import webdriver

from selenium.webdriver.common.action_chains import ActionChains

from selenium.webdriver.common.by import By


driver = webdriver.Chrome()

driver.get("https://www.google.com")


actions = ActionChains(driver)

element = driver.find_element(By.NAME, "q")
```

```
actions.move_to_element(element).click().send_keys("Hello").perform()


driver.quit()
```

Q9) Design and implement web table interaction to extract and process data from table elements.

```
from selenium import webdriver

from selenium.webdriver.common.by import By


driver = webdriver.Chrome()

driver.get("https://www.w3schools.com/html/html_tables.asp")


rows = driver.find_elements(By.XPATH, "//table[@id='customers']/tbody/tr")


for row in rows:

    print(row.text)


driver.quit()
```

Q10) Design and implement wait mechanisms to handle dynamic web elements using implicit or explicit waits.

```
from selenium import webdriver

from selenium.webdriver.common.by import By


driver = webdriver.Chrome()


driver.implicitly_wait(10)


driver.get("https://www.google.com")


search = driver.find_element(By.NAME, "q")

search.send_keys("Selenium")
```

```
driver.quit()
```